CSE 270 - Sports Analytics

Term/Year: Fall 2026
Subject Code and Course Number: CSE270
Course Title: Sports Analytics
Number of Credits: 3
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Office Location: 429W
Office Hours: Fridays: 12:20-13:30 or by appointment
Class Schedule: MWF 11:30-12:20 pm Room: 214M
Teaching Assistant: TBD
Teaching Assistant Email: TBD
Moodle Enrollment Key: Analytics

Course Description:
Professional sport organizations are using analytics to make better decisions on team formation, playing strategy etc. Enthusiasts use analytics to predict the outcome of a sporting event and to try to quantify reasons that lead to victory. The course will examine how different statistical and data analytics methods can be used to analyze game-day (in-play) sports data and for pre- and post-game sports performance modelling. We will focus on several team games, e.g. soccer, basketball, American football and baseball. The course will use a statistical programming language such as R and assessment may include problem sets, individual or group projects and written examinations.

Always bring a laptop with you !

Prerequisites: *BUS110 OR CS108 OR CSE241*

Co-Requisites: ***None***

Recommended readings:

1. Winston, W. L. (2012). Mathletics: how gamblers, managers, and sports enthusiasts use mathematics in baseball, basketball, and football. Princeton, NJ: Princeton University Press.
2. Koning, R. H., & Albert, J. (2008). Statistical thinking in sports. Boca Raton, FL: Chapman & Hall/CRC
3. Lewis, M. (2003). Moneyball: The art of winning an unfair game. W.W. Norton & Company

Course Syllabus is subject to change to address student needs.

	Topic	Reading	Non-Reading Home Tasks/Due Dates for Assignments
1	Introduction to Sports Analytics and R programming		
2	Predicting the game result with distributions		
3	Predicting the game result with models		
4	Winning percentage/Elo ratings		
5	Skills and Luck in sports/Competitive Balance		
6	Adjusted plus minus		

7	Individual performance analysis in sports		
8	Rash Model		
9	Spatial Analytics in Sports		
10	Special Topics and advanced metrics in Sports Analytics		

Student Learning Outcomes:

Program Goal	***Program Student Learning Outcomes*** *Students will be able to:*	***Course-based Student Learning Outcomes*** *In this course, students will be able to:*
Goal 4: Develop articulate, conscientious leaders and problem solvers who are committed to contributing to their fields and society.	*4.2 Think critically and creatively, conceptualizing real-world problems from different perspectives.*	*-Use statistical methods and tools to analyse sport events* *-Identify sport related problem and statistical tools appropriate to solve it*
Goal 5: Provide students with a broad foundation of knowledge and skills and cultivate a commitment to life-long learning	*5.1 Use common software and information technology to pursue inquiry relevant to their academic and professional fields, and personal interests.*	*-Use R programming language to solve different statistical problems*
	5.2 Weigh evidence and arguments, and appreciate and engage in diverse modes of inquiry characteristic of historical, cultural, political, economic, and quantitative disciplines.	*-Use critical thinking to analyse causes of events in sport* *-Discover new knowledge in sports using statistical tools and models*

Course Structure
The instructor-led class will meet 3 times per week.

Method of Evaluation:

Homeworks	10%
Exam	20%
Group Project	40%
In class quiz	10%
Participation	15%
Attendance	5%

Grades

A+	% ≥ 95.0
A	93.0 ≤ % < 95.0
A-	90.0 ≤ % < 93.0
B+	86.66 ≤% < 90.0
B	83.33 ≤% < 86.66
B-	80.0 ≤ % < 83.33
C+	76.66 ≤ %< 80.0
C	73.33 ≤ % < 76.66
C-	70.0 ≤ % < 73.33
D+	66.66 ≤ % < 70.00
D	63.33 ≤ % < 66.66
D-	60.0 ≤ % < 63.33
F	% < 60.0

A brief explanation of the weighted components:

Homeworks:

Tasks from the topics covered during the weeks. R markdown report should be reported.

Exam:

Paper based or hybrid exam will be conducted near the end of the class. More details will be provided during the course.

Group Project:

Groups of 4 or 5 students will identify a research question and define a hypothesis to be tested.

Attendance:

Physical attendance of students, checked in the beginning of the class, counted as a percentage.

Participation:

The students will have different small individual projects and will present their findings throughout

Late Policy - Assignments

20% of the score will be deducted from an assignment each day that it is late, assignments submitted later than 2 days, won't be accepted.

Make-up Procedures

Students must submit convincing evidence of a medical or other emergency that makes completing an assignment or taking an exam or quiz at the scheduled time impossible.

Communication: [Note: In this section, provide students with guidelines on how you will communicate with them and how you expect students to communicate with you. For example, will you set up a discussion forum in Moodle for course-related Q&A (and will you encourage students to use that forum for questions that others in the class might benefit from)? How will you use email? How do you expect students to use email? Where will you post important information?]

Policy on Grade Appeal
Students are entitled to appeal grades in line with the university's *Grade Policies* policy which is available online at http://policies.aua.am/policy/11

Standards for Academic Integrity
Students are required to conduct themselves in an academically responsible and ethical manner in line with AUA's *Student Code of Ethics.* Acts of academic dishonesty impair the academic integrity of AUA and create an unfair academic advantage for the student involved and other member(s) of the academic community. These acts are subject to disciplinary measures as prescribed in the AUA Student Code of Ethics, http://policies.aua.am/policy/10

The Student Code of Conduct can be found at http://policies.aua.am/policy/101

Special Needs:
Students requesting special accommodations for learning should contact the Office of Student Affairs, studentaffairs@aua.am, https://studentaffairs.aua.am/disability-support-services/ by the end of the Add/Drop period with such requests.